FAIR DATA – SESSION 2

UGent Data Stewards

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CC BY

OUTLINE

- Intro
  - Why do we need FAIR principles?
  - Data concepts

- Creating documentation, metadata, provenance
- Using vocabularies, standards, formats
- Citing and reusing data through identifiers, access rights, licenses
- FAIR at UGent
- Example repository

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POLL

Did you attend FAIR data – session 1?

- Yes
- No

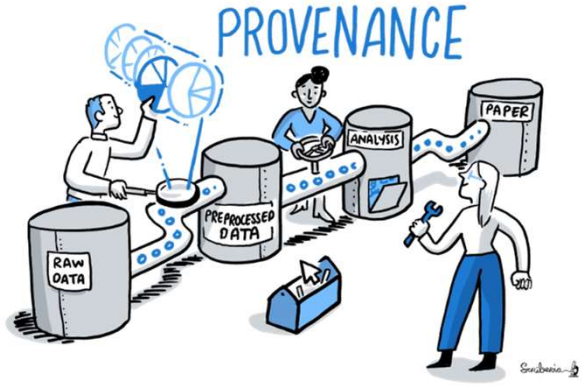
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WHY DO WE NEED FAIR PRINCIPLES?

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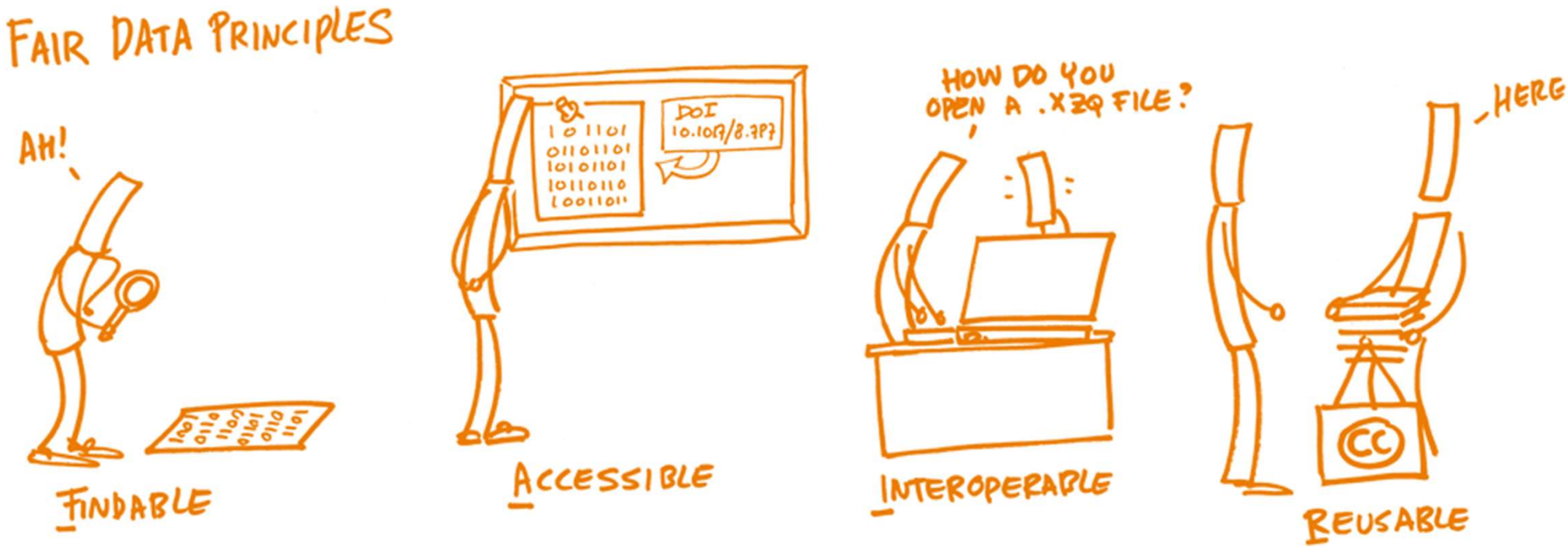
RATIONALE FOR DEVELOPING FAIR PRINCIPLES

- Urgent need to improve the infrastructure supporting the reuse of research data
- Specific emphasis on enhancing the ability of machines to automatically find and use the data, in addition to supporting its reuse by individuals
- The FAIR principles are a guideline to enhance the data reusability



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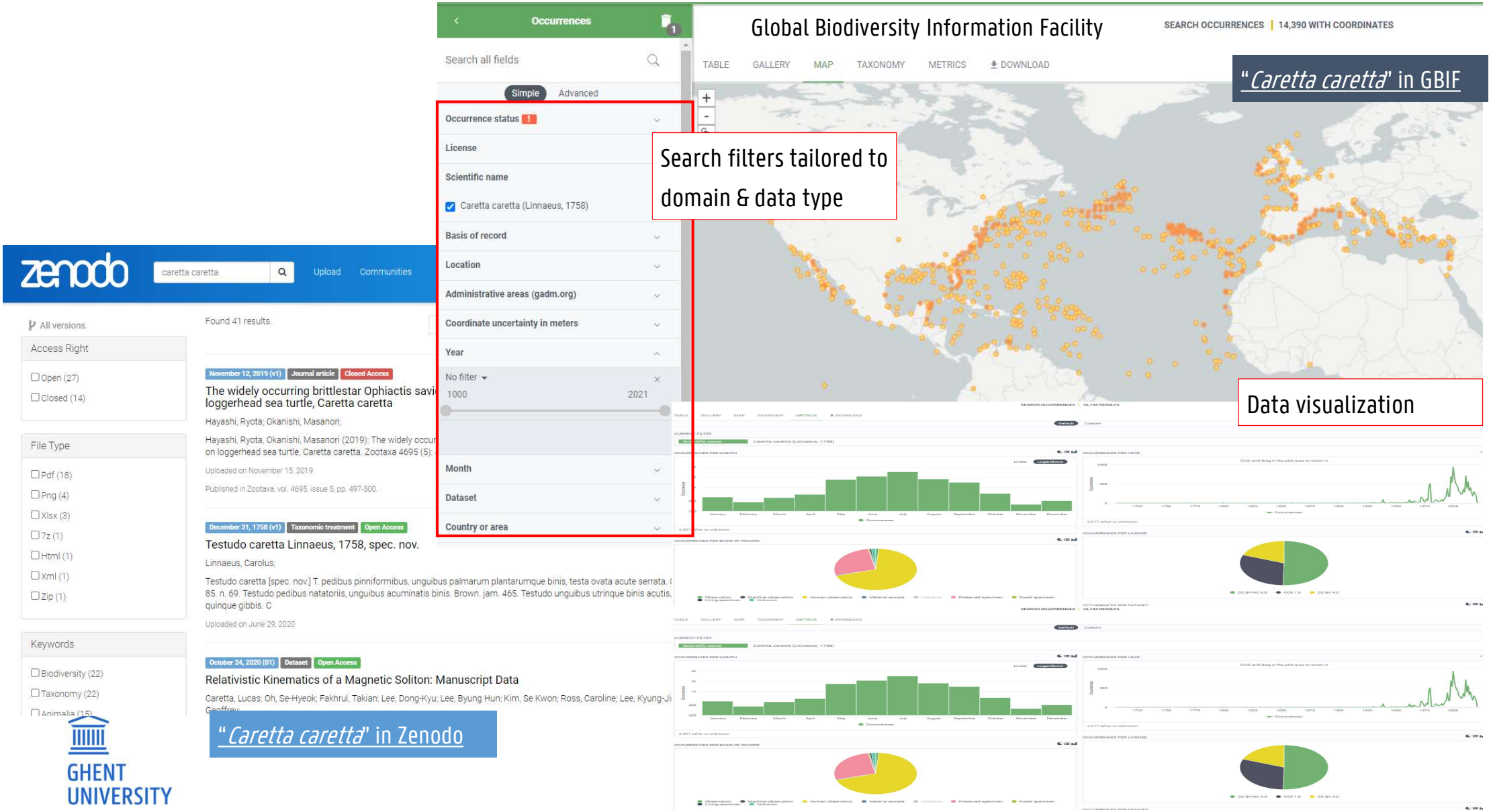
Wilkinson, M., Dumontier, M., Aalbersberg, I. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3,** 160018 (2016). https://doi.org/10.1038/sdata.2016.18



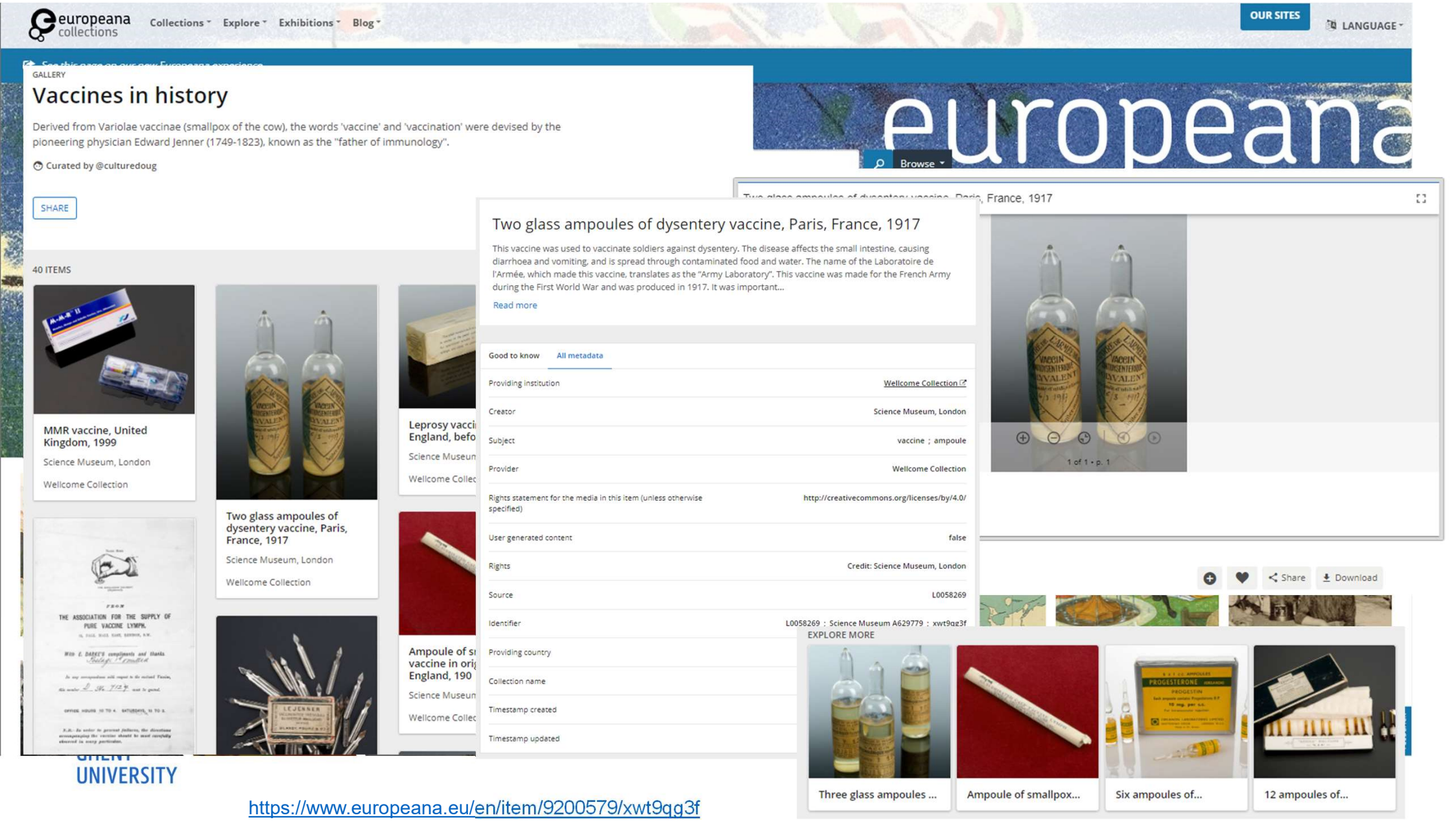
https://book.fosteropenscience.eu/

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Wilkinson, M., Dumontier, M., Aalbersberg, I. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3,** 160018 (2016). https://doi.org/10.1038/sdata.2016.18



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https://www.europeana.eu/en/item/9200579/xwt9qg3f

CROSS-DISCIPLINARY DATA INTEGRATION TO TACKLE GLOBAL CHALLENGES

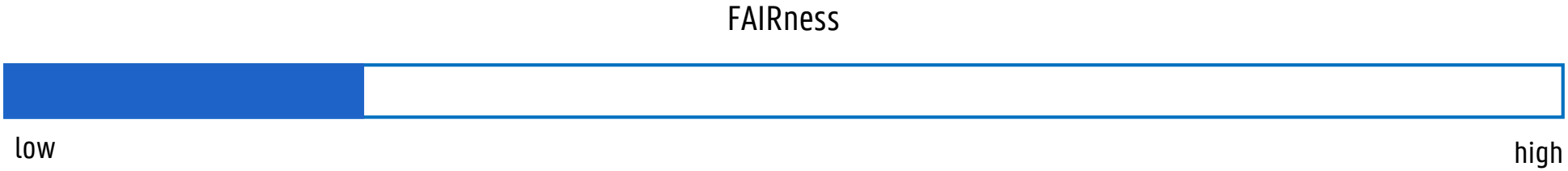


ENVRI-FAIR

Presentation Andreas Petzold at EOSC symposium 2021

FAIRNESS: DEGREE TO WHICH FAIR PRINCIPLIES ARE APPLIED

- FAIR principles provide ‘steps along a path’ toward machine-actionability
- Adopting the FAIR principles, in whole or in part, leads the data along the continuum towards the optimal state
- Depends on the discipline and available tools to increase FAIRness
- Goal: FAIR by design → planning



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DATA CONCEPTS

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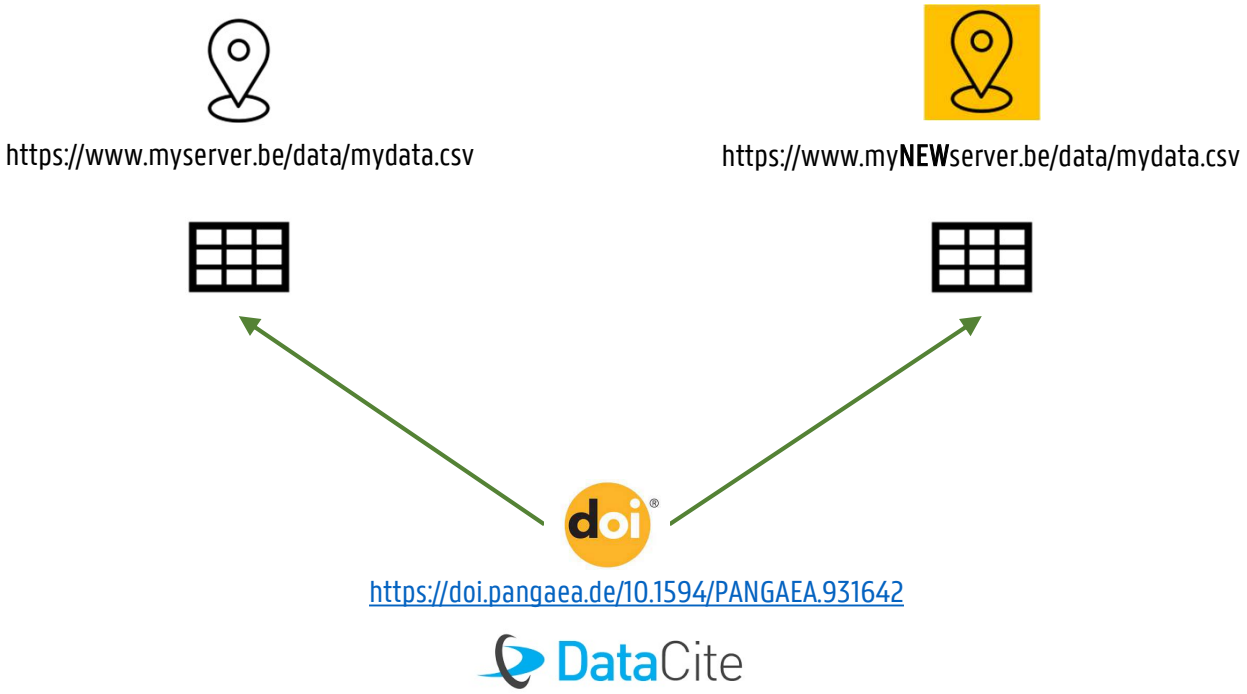
MACHINE READABILITY OR ACTIONABILITY

to allow for the scope, scale and speed often needed in contemporary science

a) identify the type of object (with respect to both structure and intent)
b) determine if it is useful within the context of the agent's current task by interrogating metadata and/or data elements
c) determine if it is usable, with respect to license, consent, or other accessibility or use constraints
d) take appropriate action, in much the same manner that a human would

PERSISTENT IDENTIFIERS

DOI - digital object identifier
for identification of digital objects



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Similar to ORCID for researchers or ROR ID for organizations

PERSISTENT IDENTIFIERS

	Identifier	Globally unique identifier (guid, uuid)	Persistent identifier (PID)
What	a name that identifies a unique object gene identifier ENSG00000139618 book identifier ISBN 817525766-0	ensures that there are no two identical identifiers that point to different digital objects 123e4567-e89b-12d3-a456-426614174000	ensures that the digital object will remain findable over time and reduces the risk of broken links
How	can be (arbitrarily) assigned by anyone	by using a mechanism (e.g. using a registry service) for assigning identifiers	is determined by the commitment of the organisation that assigns and manages the identifier

Examples of globally unique, persistent identifiers

Identifier:	doi	iD	R
Organisation:	DataCite	ORCID Connecting Research and Researchers	ROR
Scope:	Digital objects	Researchers	Research organisations

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METADATA

STRUCTURED, MACHINE-READABLE DOCUMENTATION ABOUT DATA

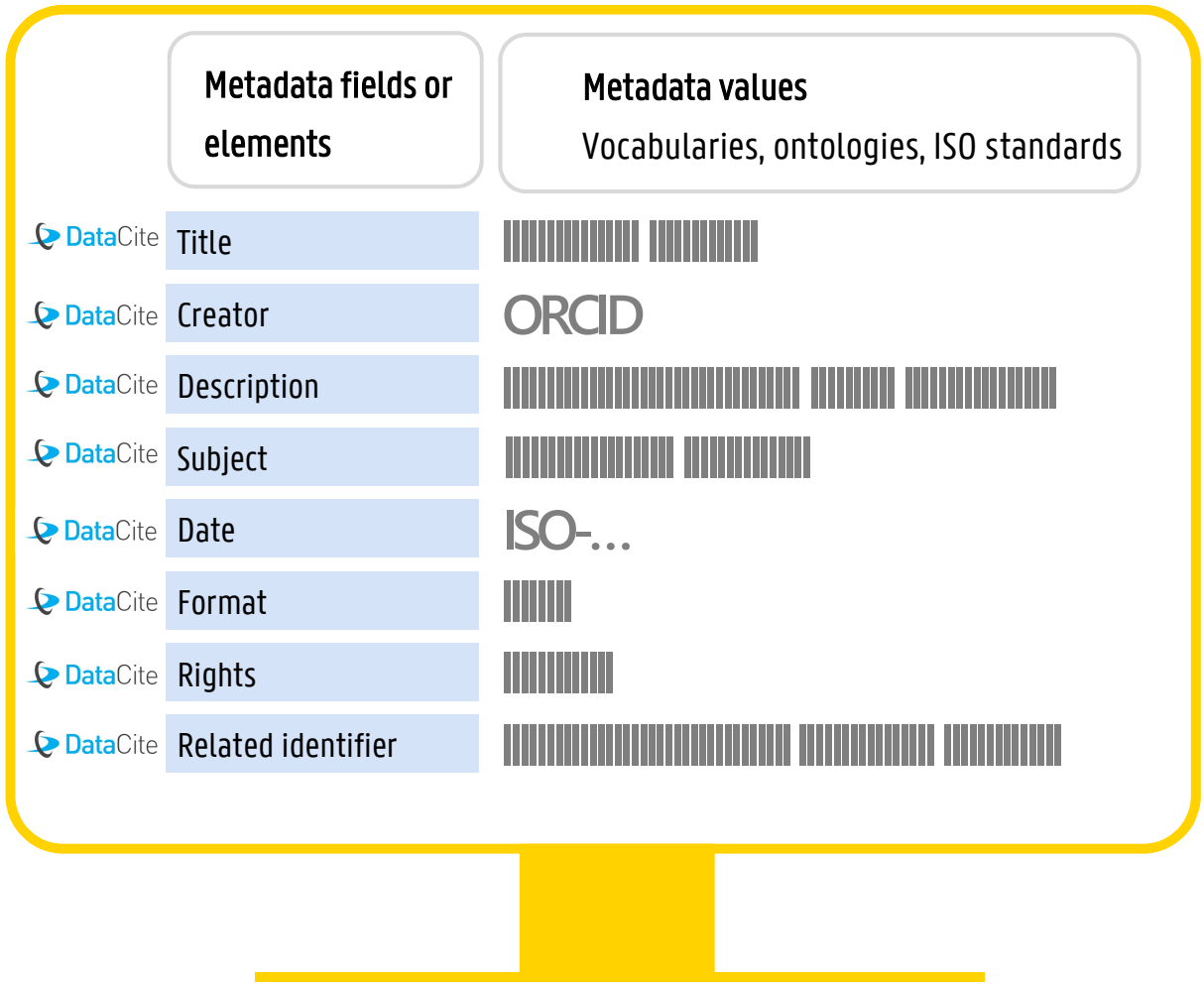


Title	
Creator	ORCID
Abstract	
Keywords	
Date	ISO-…
File format	
License	
Related publication	

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VOCABULARIES AND STANDARDS

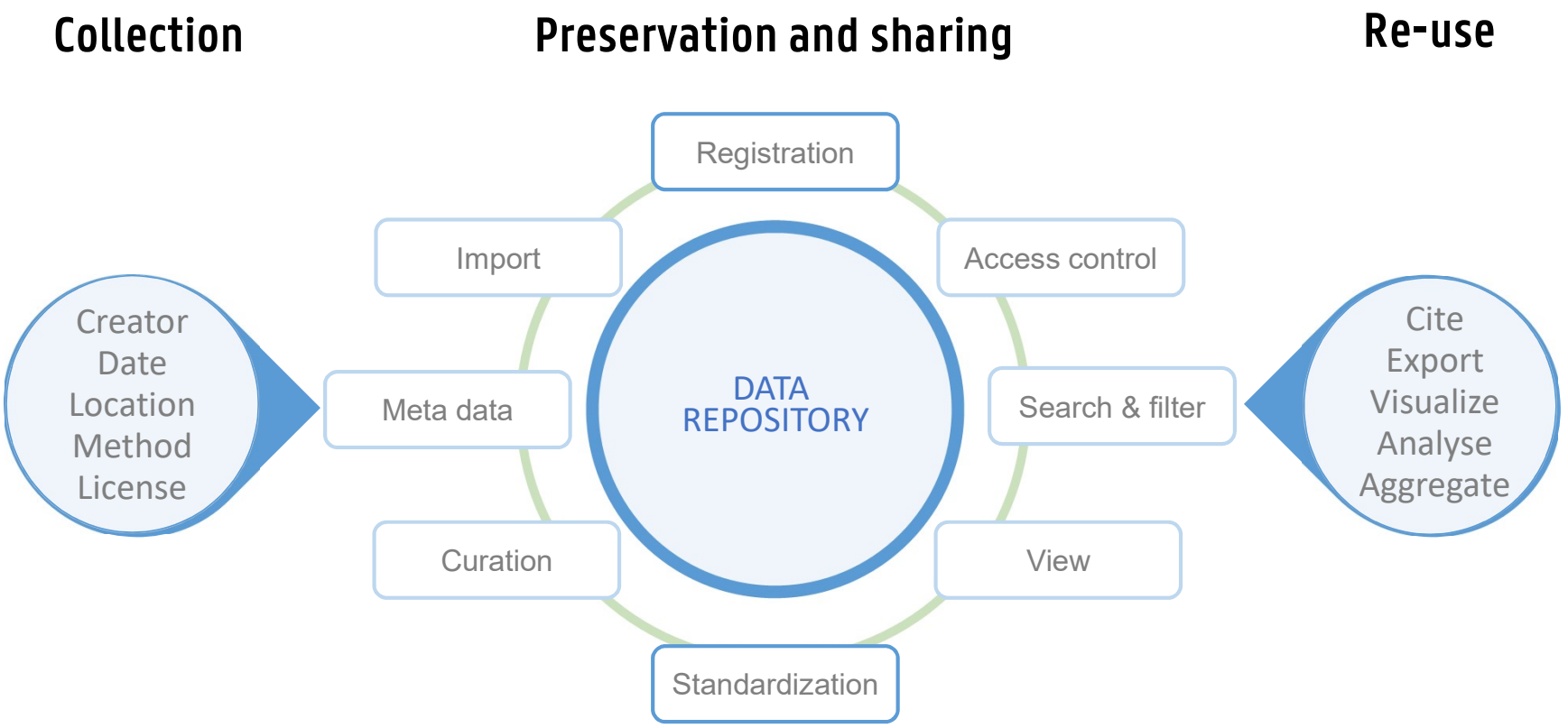
STANDARDIZED METADATA



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DATA REPOSITORY

A database infrastructure that collects, manages, and stores data sets for data preservation and sharing



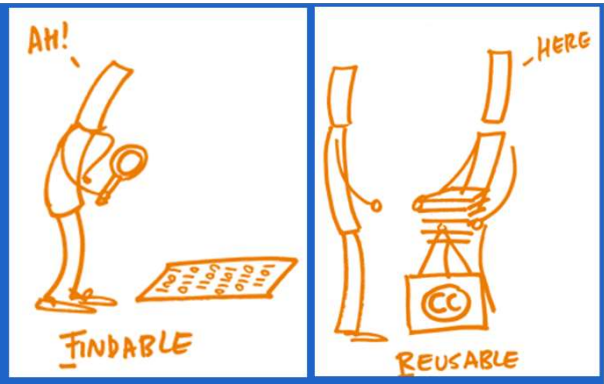
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POLL

Do you know which identifiers, repositories, standards are common in your field?

- No, I do not know
- I know which identifiers are used
- I know which repositories are used
- I know which standards are used

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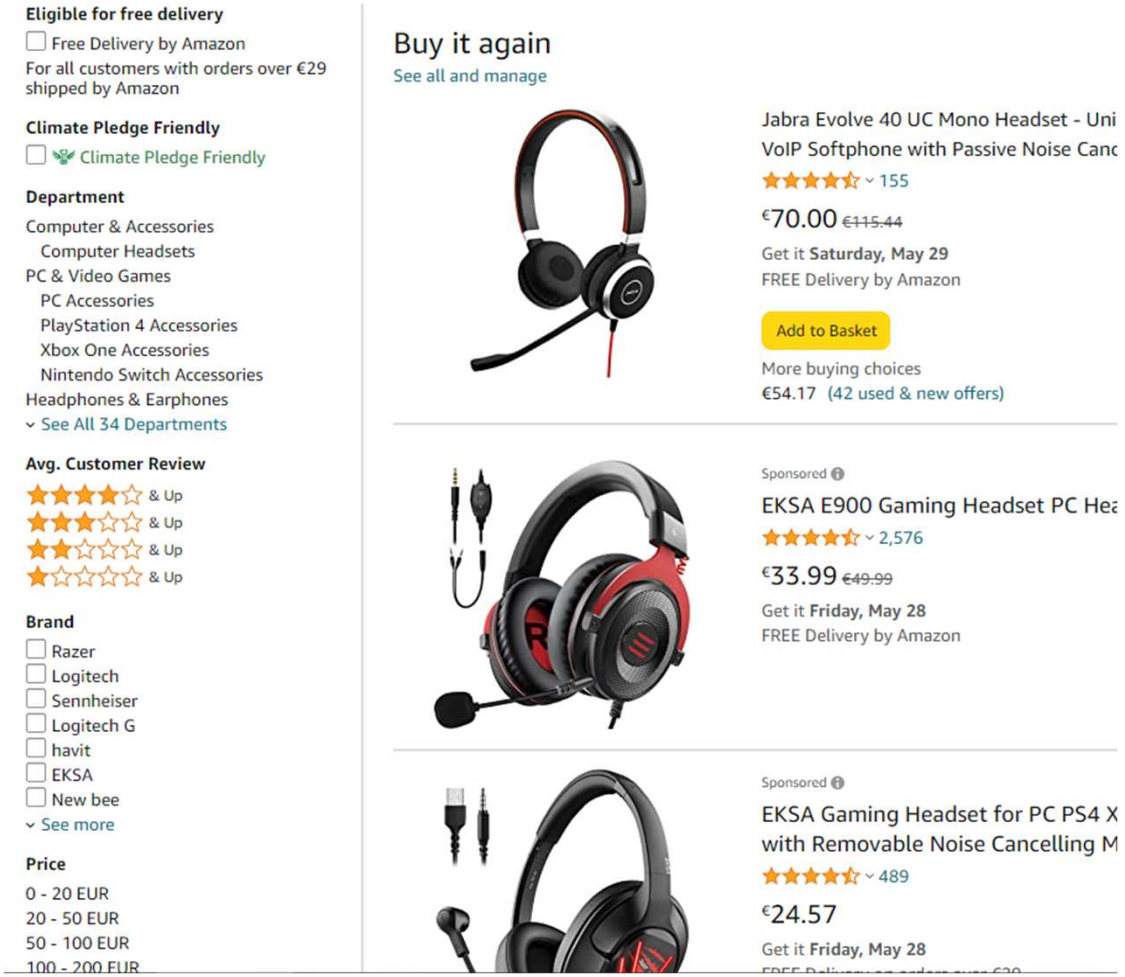
CREATING DOCUMENTATION, METADATA, PROVENANCE

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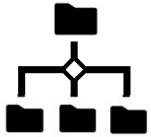
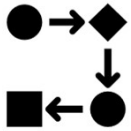
TAGGING AND FILTERING







LEVELS OF DATA DOCUMENTATION

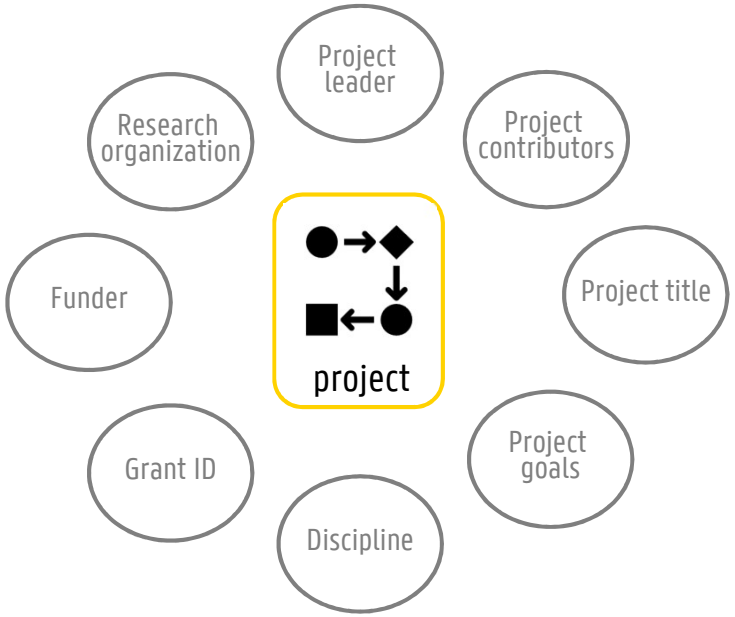
DOCUMENTATION LEVEL	DOCUMENTATION FORMATS	DOCUMENTATION CONTENT
 Item Variable	Codebook, data dictionary	Methods, units, possible values, synonyms, abbreviations
 Sample Study object	Table, protocol	Study object, time, place
 Dataset	Readme, protocol	Methods, technologies, format, who, what, why, where, when
 Project	Notebook, document, protocol, readme, data overview table	Subject, grant, contributors, methods, process, workflow

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PROJECT LEVEL DOCUMENTATION

README, PROTOCOL

WHAT



HOW

Cornell

AUTHOR_DATASET_ReadmeTemplate.txt

This DATSETNAMEreadme.txt file was generated on YYYY-MM-DD by NAME
<help text is included in angle brackets, and can be deleted before saving>

GENERAL INFORMATION

1. Title of Dataset:

2. Author Information
   A. Principal Investigator Contact Information
      Name:
      Institution:
      Address:
      Email:

   B. Associate or Co-investigator Contact Information
      Name:
      Institution:
      Address:
      Email:

   C. Alternate Contact Information
      Name:
      Institution:
      Address:
      Email:

3. Date of data collection (single date, range, approximate date) <suggested format YYYY-MM-DD>:

4. Geographic location of data collection <latitude, longiute, or city/region, State, Country, as appropriate>:

5. Information about funding sources that supported the collection of the data:

SHARING/ACCESS INFORMATION

1. Licenses/restrictions placed on the data:

2. Links to publications that cite or use the data:

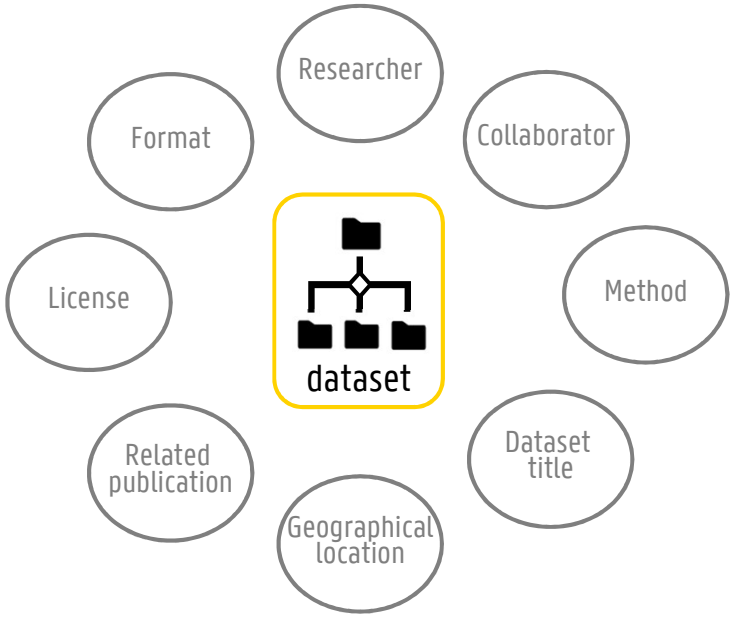
3. Links to other publicly accessible locations of the data:

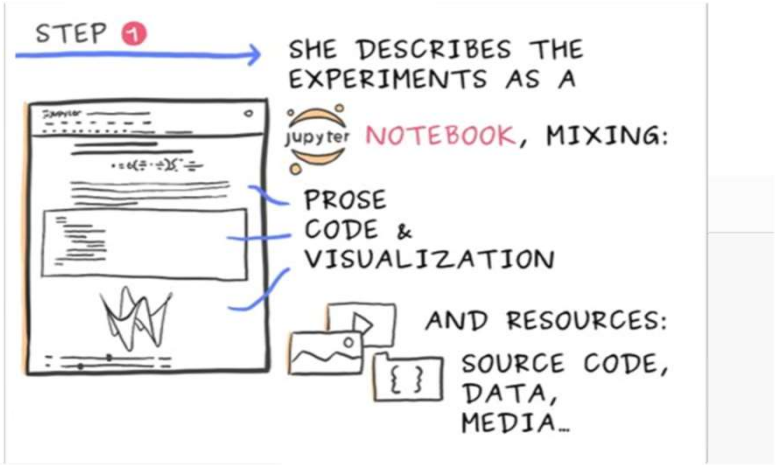
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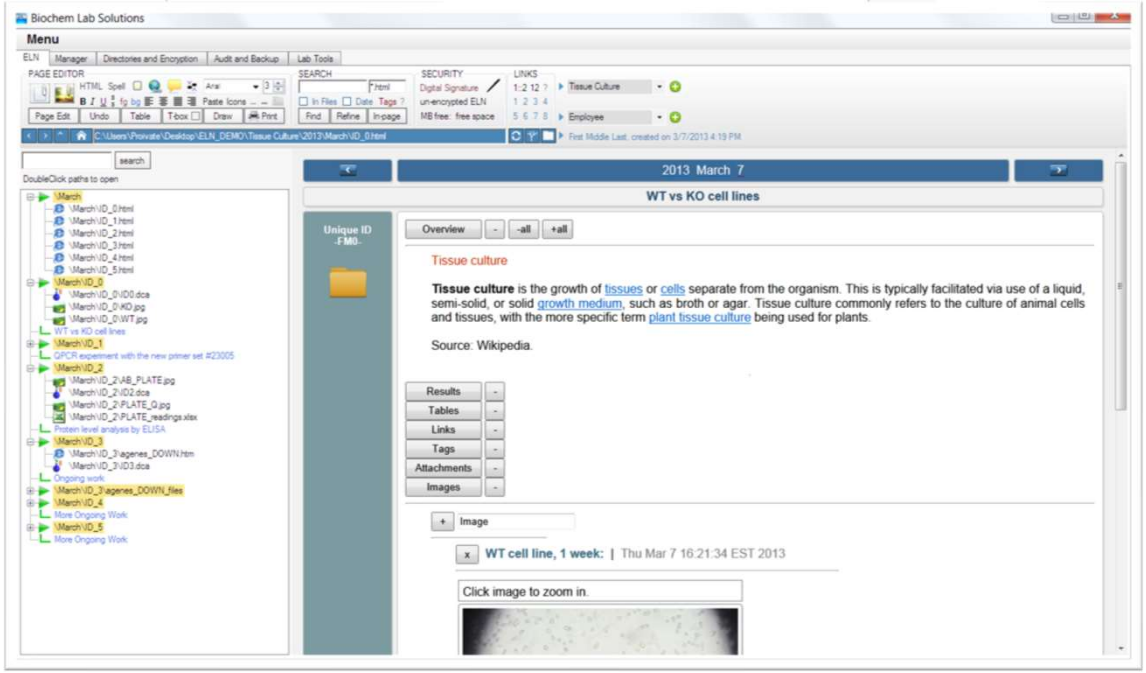
DATASET LEVEL DOCUMENTATION

README FILE, NOTEBOOK

WHAT





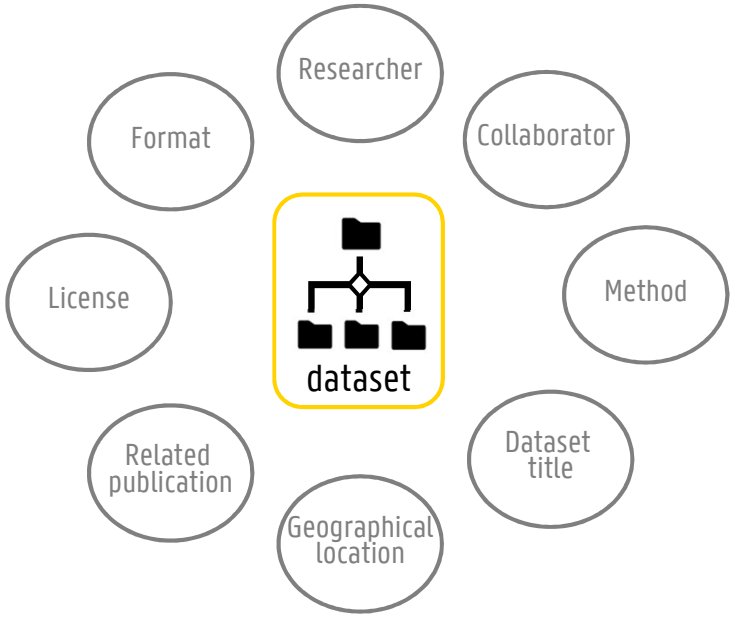


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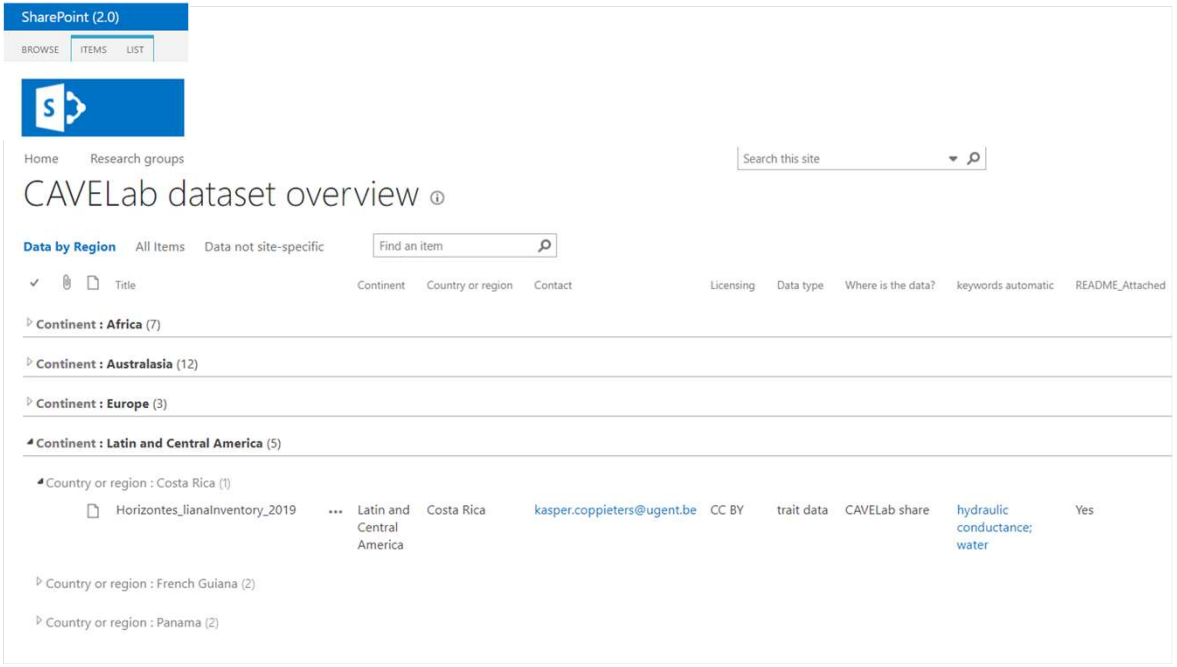
DATASET LEVEL DOCUMENTATION

METADATA ON SHAREPOINT

WHAT



HOW



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Home Research groups

Search this site

CAVELab dataset overview

Data by Region All Items Data not site-specific

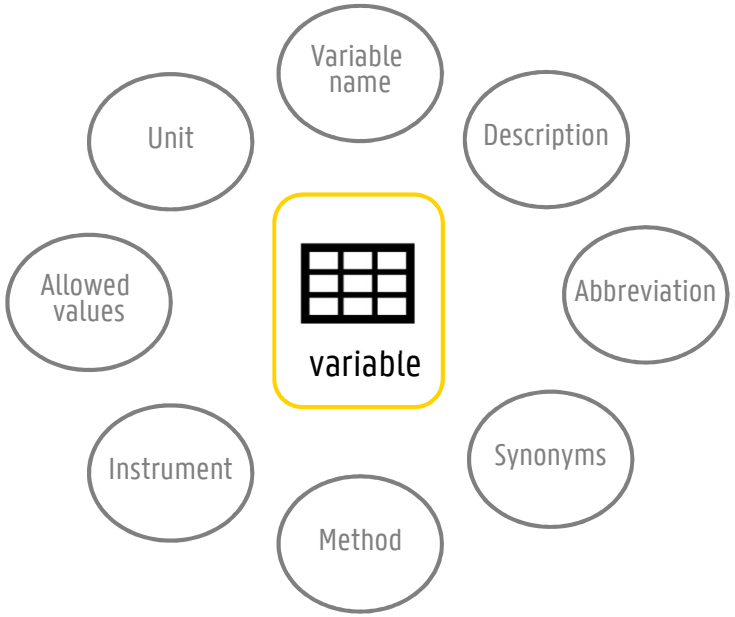
Find an item

	Title	Continent	Country or region	Contact	Licensing	Data type	Where is the data?	keywords automatic	README_Attached
▷ **Continent : Africa** (7)									
▷ **Continent : Australasia** (12)									
▷ **Continent : Europe** (3)									
◢ **Continent : Latin and Central America** (5)									
◢ Country or region : Costa Rica (1)									
	Horizontes_lianaInventory_2019 •••	Latin and Central America	Costa Rica	kasper.coppieters@ugent.be	CC BY	trait data	CAVELab share	hydraulic conductance; water	Yes
▷ Country or region : French Guiana (2)									
▷ Country or region : Panama (2)									

VARIABLE LEVEL DOCUMENTATION

CODEBOOK OR DATA DICTIONARY

WHAT



HOW

Sheet_1

Show rows with cells including:

Variable	Variable name	Mesaurement unit	Allowed values	Description
Participant ID number	ID	Numeric	001-999	ID number assigned to participant in sequential order
Group number	GROUP	Numeric	1-30	Group assigned to participant based on ID number
Age in years	AGE	Numeric	18.0-65.0	Age of participant in years
Date of birth	DOB	mm/dd/yyyy	1-12/1-31/1951-1998	Participant's date of birth
Gender	SEX	Numeric	1 = male 2 = female	Participant's gender
Date of survey	SURVEY	mm/dd/yyyy	01/01/2015 – 01/01/2016	When the participant completed the survey
Self-reported consumer spending	SPEND	Numeric	0-100,000,000	Self-reported average yearly expenditure
Market sentiment	SENTIMENT	Numeric	1 = negative 2 = neutral 3 = positive	Sentiment towards US domestic economy
Actual GDP growth	GDP	Numeric	-5.0-5.0	Average US yearly GDP growth

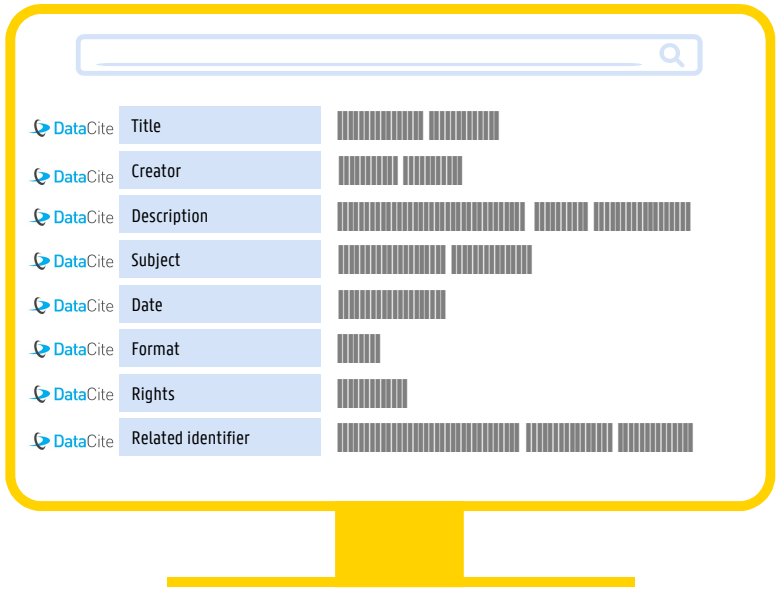
How to Make a Data Dictionary

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METADATA STANDARDS FOR DATASETS

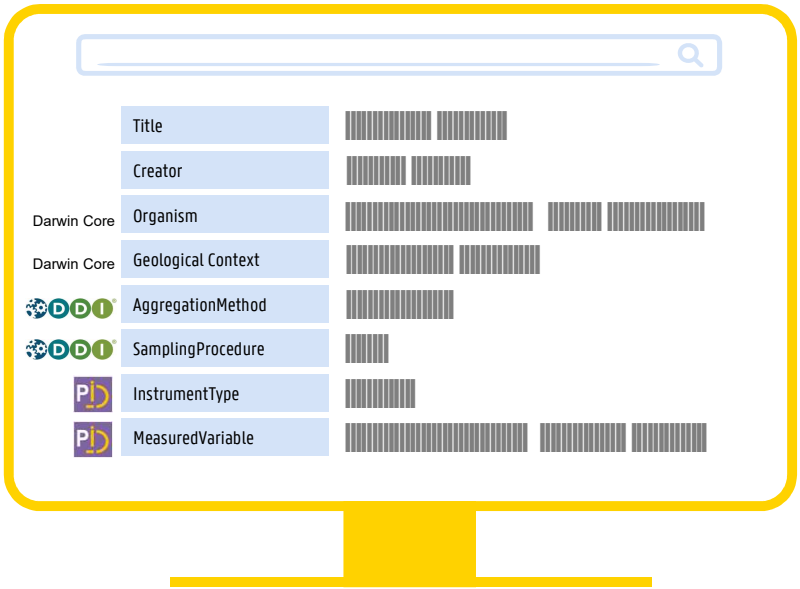
Domain-agnostic standards:

DataCite, Dublin Core, schema.org, etc.



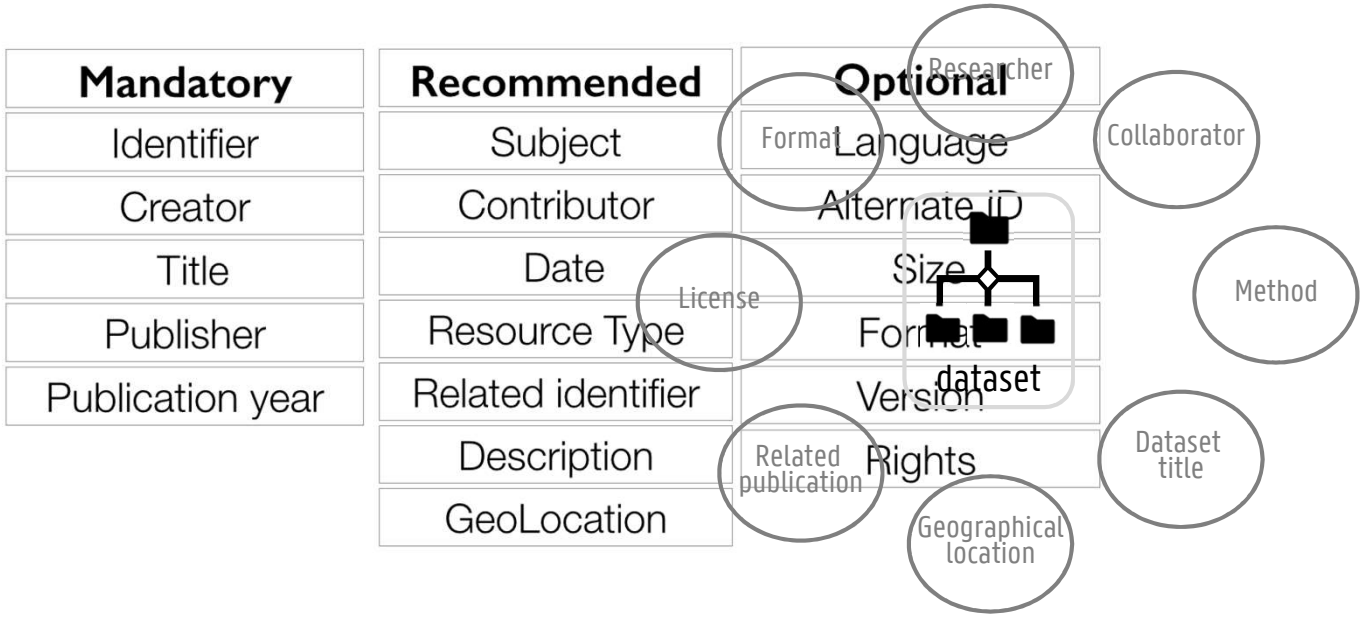
Domain-specific standards:

Darwin Core, Data Documentation Initiative (DDI), Ecological Metadata Language, Persistent Identification of Instruments (PIDINST), etc.



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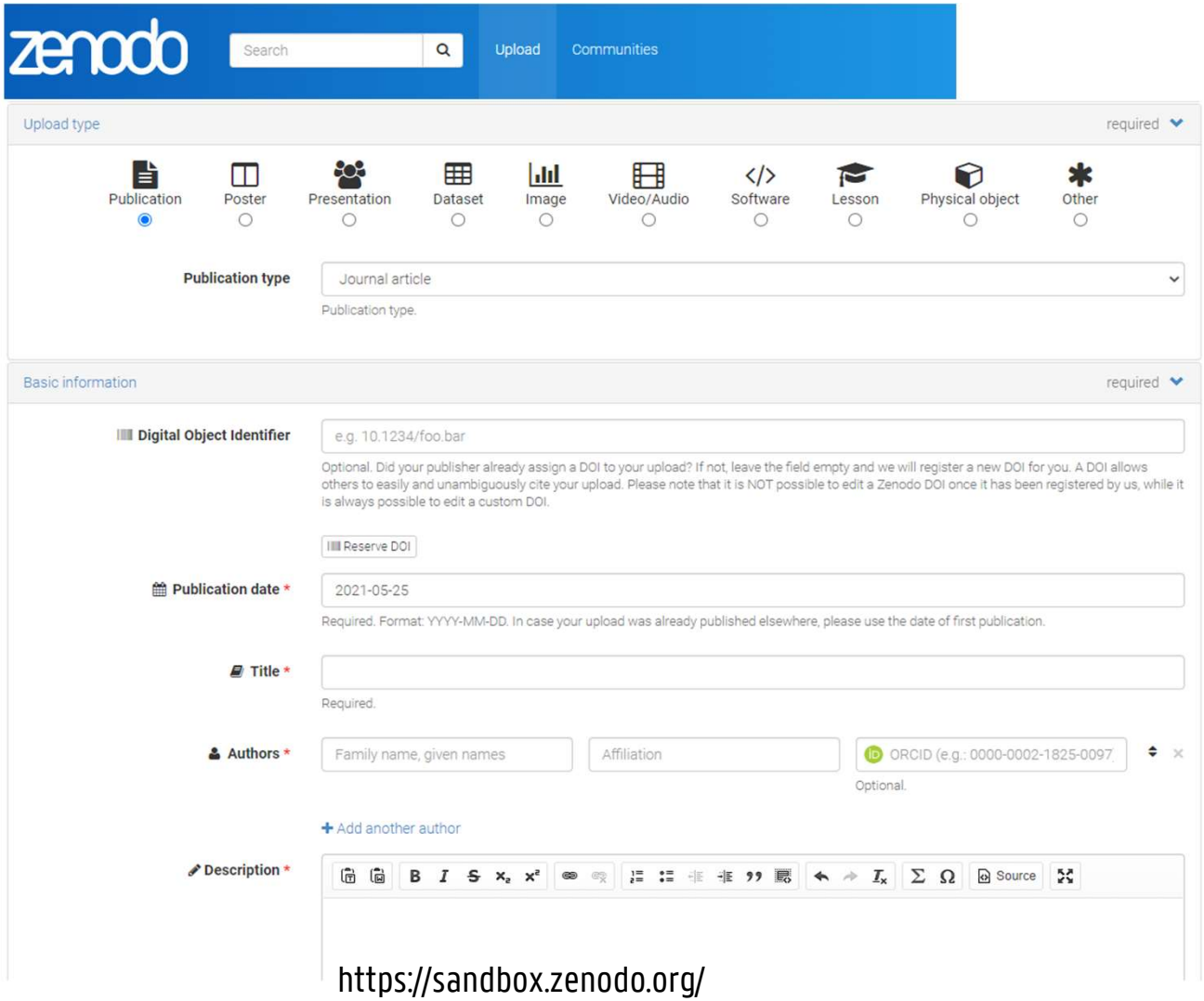
GENERATE A METADATA TEMPLATE



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How to: https://fairplus.github.io/the-fair-cookbook/content/recipes/interoperability/creating-minimal-metadata-profiles.html#how-to-generate-a-metadata-template
Example: https://fairplus.github.io/the-fair-cookbook/content/recipes/interoperability/transcriptomics-metadata.html

METADATA FORM OF REPOSITORY



https://sandbox.zenodo.org/

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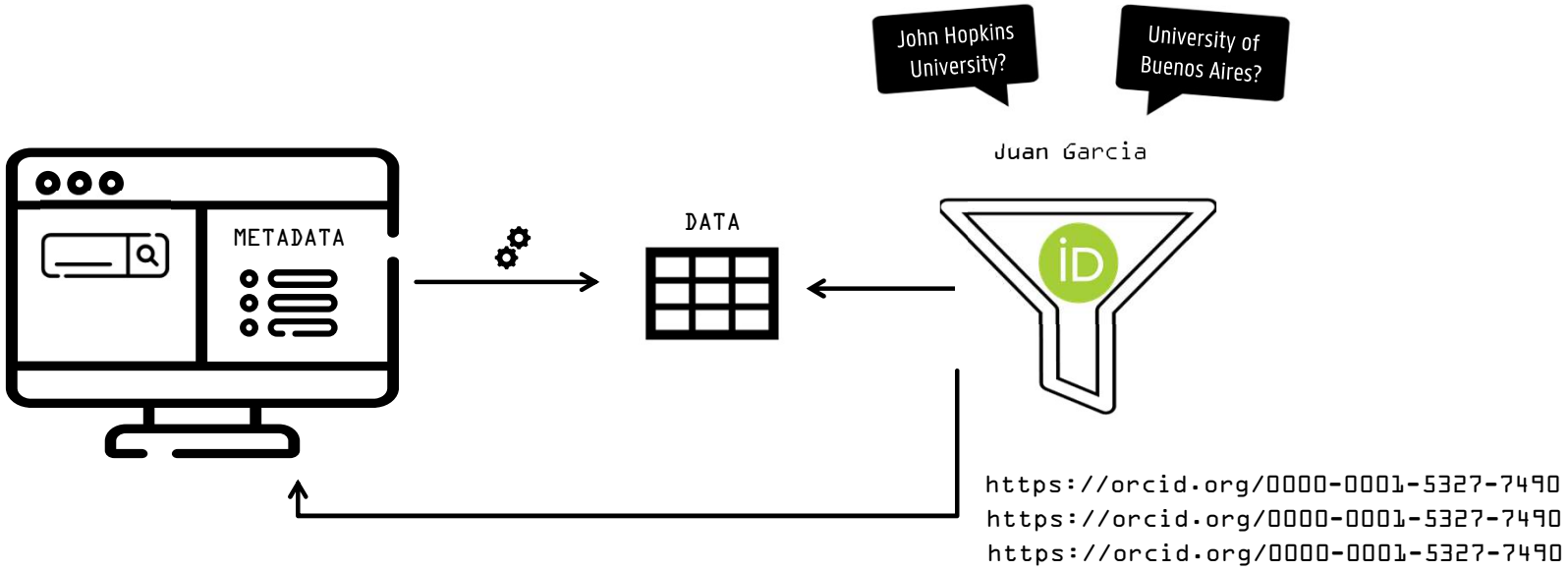




USING VOCABULARIES, STANDARDS, FORMATS

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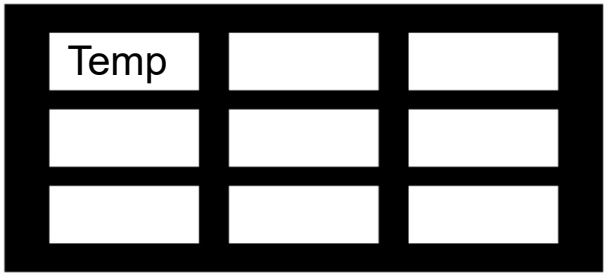
CONTROLLED VOCABULARIES TO STANDARDIZE TERMINOLOGY



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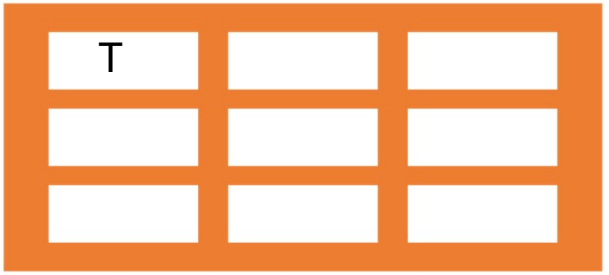
LACK OF CONTROLLED VOCABULARY

Tedious matching, mapping of terms



Researcher 1





Researcher 2

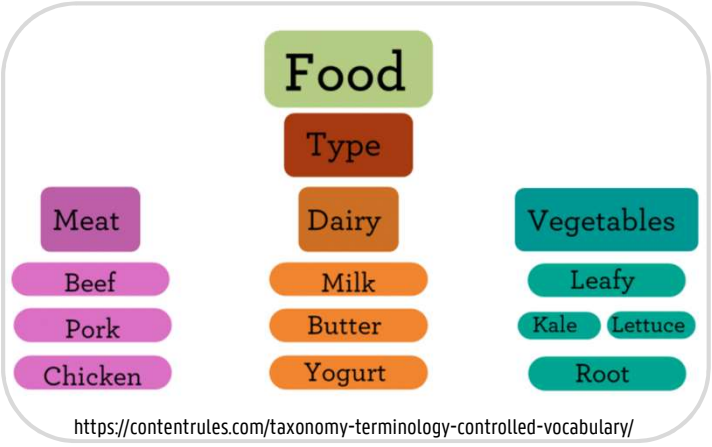
Concept	LABEL	URL
Temperature of the atmosphere by psychrometer dry bulb	CDTAPS01	http://vocab.nerc.ac.uk/collection/P01/current/CDTAPS01/
Temperature of the atmosphere by dry bulb thermometer	CDTASS01	http://vocab.nerc.ac.uk/collection/P01/current/CDTASS01/
Temperature of the atmosphere by thermometer	CDTAZZ01	http://vocab.nerc.ac.uk/collection/P01/current/CDTAZZ01/
Temperature of the atmosphere by tide gauge ambient temperature sensor	CTLGPR01	http://vocab.nerc.ac.uk/collection/P01/current/CTLGPR01/

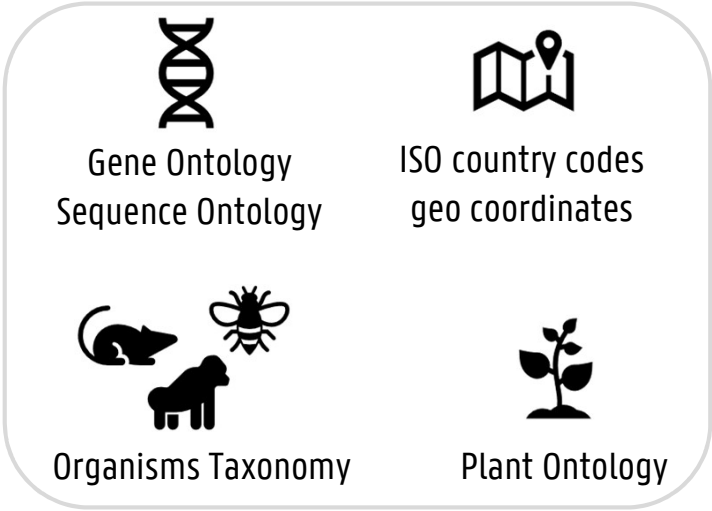
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STANDARD OR CONTROLLED VOCABULARIES

RESULT OF CONSENSUS TERMINOLOGY ACROSS COMMUNITY

- Deal with the “meaning”: semantic standardization
- Provide consistency, avoid ambiguity
- Enable machine readability by using PIDs for terms





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USE STANDARD (OPEN) FILE FORMATS



Domain norms & software



Previous work & colleagues



Hardware & instrumentation



Facilitate collaboration



Vulnerability to obsolescence

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https://www.ukdataservice.ac.uk/manage-data/format/recommended-formats
https://youtu.be/kxxlQnc8u1I

LACK OF COMMON DATA STRUCTURE, MODELS OR FORMATS

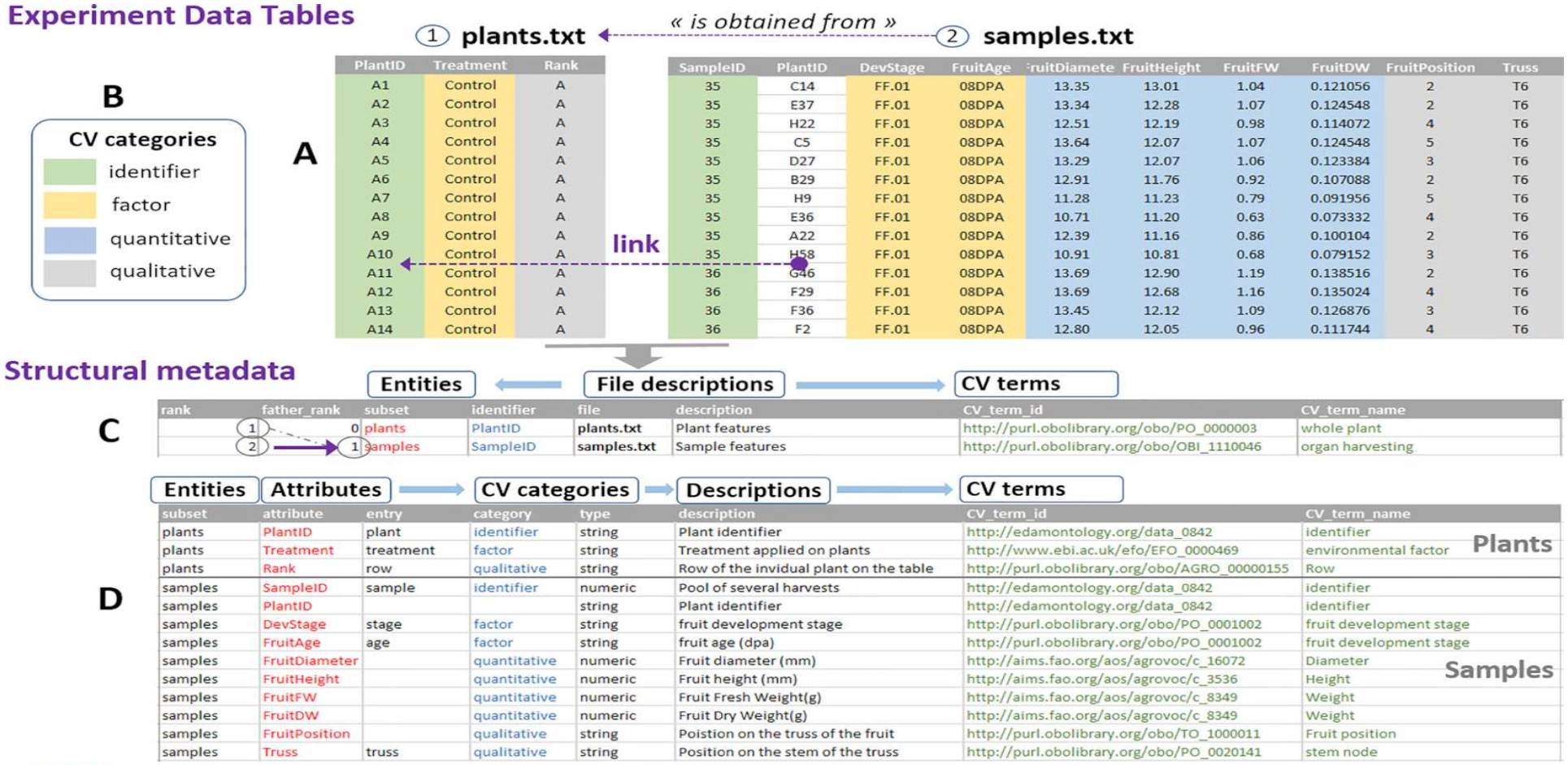
Plot: 2			
Date collecte	Species	Sex	Weight
1/8/14	NA		
1/8/14	DM	M	44
1/8/14	DM	M	38
1/8/14	OL		
1/8/14	PE	M	22
1/8/14	DM	M	38
1/8/14	DM	M	48
1/8/14	DM	M	43
1/8/14	DM	F	35
1/8/14	DM	M	43
1/8/14	DM	F	37
1/8/14	PF	F	7
1/8/14	DM	M	45
1/8/14	OT		
1/8/14	DS	M	157
1/8/14	OX		
2/18/14	NA	M	218
2/18/14	PF	F	7
2/18/14	DM	M	52
	measurement device not calibrated		

Date collecte	Species	Sex	Weight	Calibrated
1/8/14	NA			
1/8/14	DM	M	44	Y
1/8/14	DM	M	38	Y
1/8/14	OL			
1/8/14	PE	M	22	Y
1/8/14	DM	M	38	Y
1/8/14	DM	M	48	Y
1/8/14	DM	M	43	Y
1/8/14	DM	F	35	Y
1/8/14	DM	M	43	Y
1/8/14	DM	F	37	Y
1/8/14	PF	F	7	Y
1/8/14	DM	M	45	Y
1/8/14	OT			
1/8/14	DS	M	157	N
1/8/14	OX			
2/18/14	NA	M	218	N
2/18/14	PF	F	7	Y
2/18/14	DM	M	52	Y

https://datacarpentry.org/spreadsheet-ecology-lesson/02-common-mistakes/index.html

MAKING DATA TABLES MORE FAIR

VOCABULARY AND STRUCTURE COMBINED



PlantID	Treatment	Rank
A1	Control	A
A2	Control	A
A3	Control	A
A4	Control	A
A5	Control	A
A6	Control	A
A7	Control	A
A8	Control	A
A9	Control	A
A10	Control	A
A11	Control	A
A12	Control	A
A13	Control	A
A14	Control	A

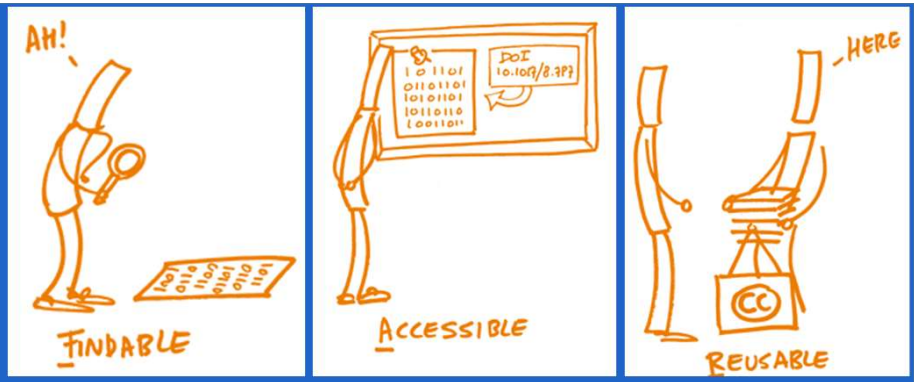
SampleID	PlantID	DevStage	FruitAge	FruitDiamete	FruitHeight	FruitFW	FruitDW	FruitPosition	Truss
35	C14	FF.01	08DPA	13.35	13.01	1.04	0.121056	2	T6
35	E37	FF.01	08DPA	13.34	12.28	1.07	0.124548	2	T6
35	H22	FF.01	08DPA	12.51	12.19	0.98	0.114072	4	T6
35	C5	FF.01	08DPA	13.64	12.07	1.07	0.124548	5	T6
35	D27	FF.01	08DPA	13.29	12.07	1.06	0.123384	3	T6
35	B29	FF.01	08DPA	12.91	11.76	0.92	0.107088	2	T6
35	H9	FF.01	08DPA	11.28	11.23	0.79	0.091956	5	T6
35	E36	FF.01	08DPA	10.71	11.20	0.63	0.073332	4	T6
35	A22	FF.01	08DPA	12.39	11.16	0.86	0.100104	2	T6
35	H58	FF.01	08DPA	10.91	10.81	0.68	0.079152	3	T6
36	G46	FF.01	08DPA	13.69	12.90	1.19	0.138516	2	T6
36	F29	FF.01	08DPA	13.69	12.68	1.16	0.135024	4	T6
36	F36	FF.01	08DPA	13.45	12.12	1.09	0.126876	3	T6
36	F2	FF.01	08DPA	12.80	12.05	0.96	0.111744	4	T6

rank	father_rank	subset	identifier	file	description	CV_term_id	CV_term_name
1	0	plants	PlantID	plants.txt	Plant features	http://purl.obolibrary.org/obo/PO_0000003	whole plant
2	1	samples	SampleID	samples.txt	Sample features	http://purl.obolibrary.org/obo/OBI_1110046	organ harvesting

subset	attribute	entry	category	type	description	CV_term_id	CV_term_name
plants	PlantID	plant	identifier	string	Plant identifier	http://edamontology.org/data_0842	identifier
plants	Treatment	treatment	factor	string	Treatment applied on plants	http://www.ebi.ac.uk/efo/EFO_0000469	environmental factor
plants	Rank	row	qualitative	string	Row of the invidual plant on the table	http://purl.obolibrary.org/obo/AGRO_00000155	Row
samples	SampleID	sample	identifier	numeric	Pool of several harvests	http://edamontology.org/data_0842	identifier
samples	PlantID			string	Plant identifier	http://edamontology.org/data_0842	identifier
samples	DevStage	stage	factor	string	fruit development stage	http://purl.obolibrary.org/obo/PO_0001002	fruit development stage
samples	FruitAge	age	factor	string	fruit age (dpa)	http://purl.obolibrary.org/obo/PO_0001002	fruit development stage
samples	FruitDiameter		quantitative	numeric	Fruit diameter (mm)	http://aims.fao.org/aos/agrovoc/c_16072	Diameter
samples	FruitHeight		quantitative	numeric	Fruit height (mm)	http://aims.fao.org/aos/agrovoc/c_3536	Height
samples	FruitFW		quantitative	numeric	Fruit Fresh Weight(g)	http://aims.fao.org/aos/agrovoc/c_8349	Weight
samples	FruitDW		quantitative	numeric	Fruit Dry Weight(g)	http://aims.fao.org/aos/agrovoc/c_8349	Weight
samples	FruitPosition		qualitative	string	Poistion on the truss of the fruit	http://purl.obolibrary.org/obo/TO_1000011	Fruit position
samples	Truss	truss	qualitative	string	Position on the stem of the truss	http://purl.obolibrary.org/obo/PO_0020141	stem node

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Gigascience, Volume 9, Issue 12, December 2020, giaa144, https://doi.org/10.1093/gigascience/giaa144.



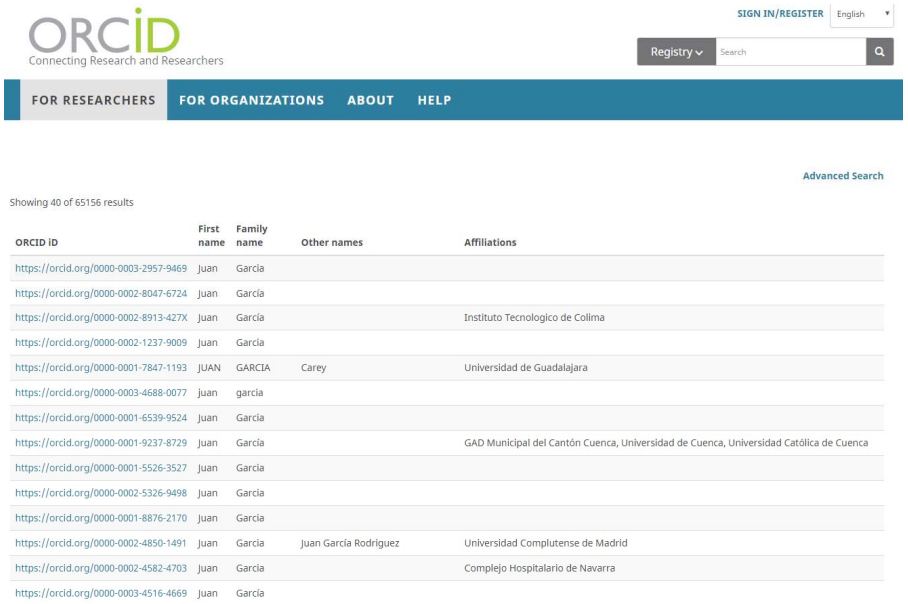
CITING AND REUSING DATA THROUGH IDENTIFIERS, ACCESS RIGHTS, LICENSES

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ORCID FOR IDENTIFICATION OF RESEARCHERS



Connection to contributions and affiliations



Unambiguous identity



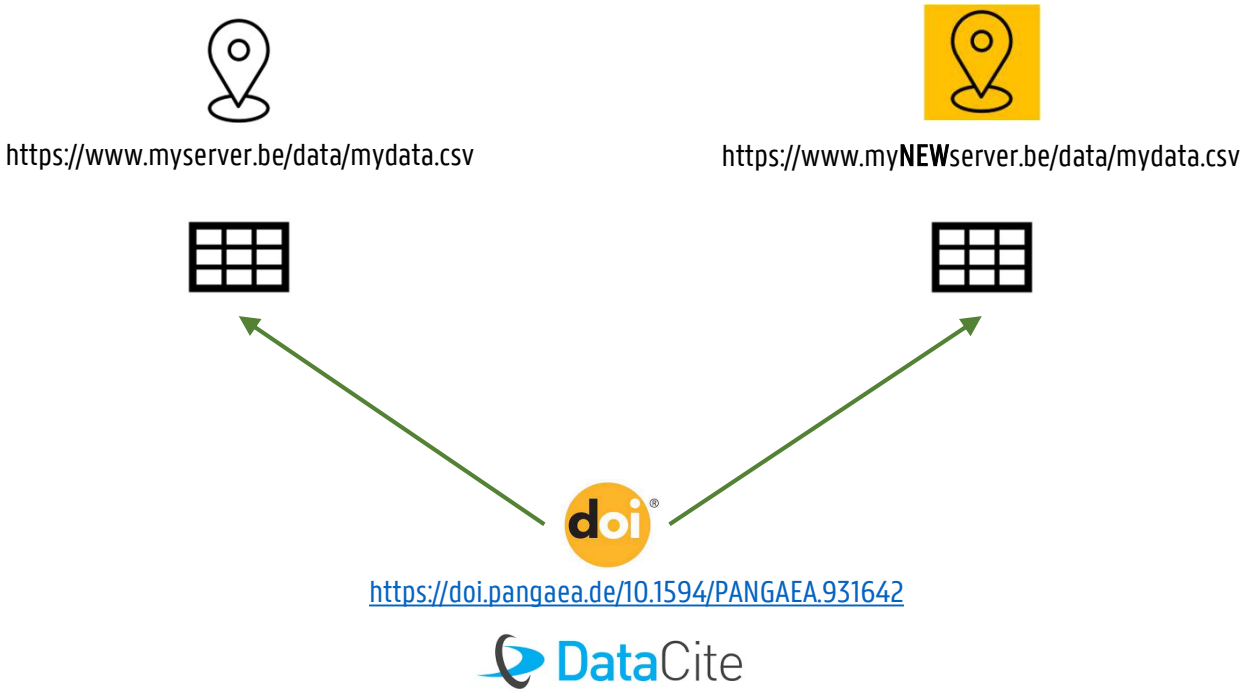
Commonly used and required

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https://orcid.org/
https://onderzoektips.ugent.be/en/tips/00001581/

DOI FOR IDENTIFICATION OF DIGITAL OBJECTS

DOI - digital object identifier



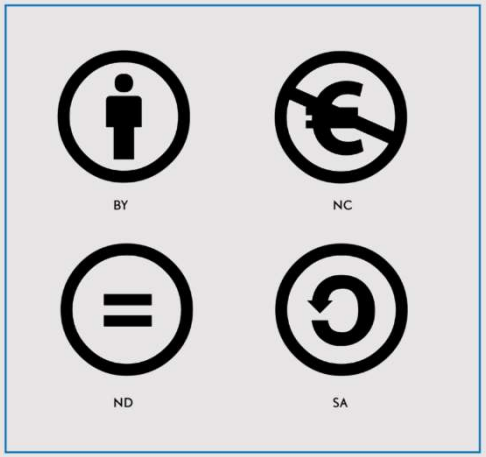
Similar to ORCID for researchers or RORID for organizations

DATACITE METADATA SCHEMA

Mandatory	Recommended	Optional
Identifier	Subject	Language
Creator	Contributor	Alternate ID
Title	Date	Size
Publisher	Resource Type	Format
Publication year	Related identifier	Version
	Description	Rights
	GeoLocation	

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FIGURE 1: PICTOGRAMS OF THE CC LICENCE FEATURE



LICENSES

PERMISSIONS & CONDITIONS: TERMINOLOGY

CC Licenses grant permissions under certain conditions:

- Attribution: give proper credit to the original creator
  - → Stacking, long chain of derivations, manual inspection of contributions.
- Copyleft (Share Alike): license the new (derived) creations under the identical terms (same license).
  - → Can be difficult to fulfil.
- Non-commerciality: to prevent the new (derived) work to be commercially exploited.
  - → Subject to interpretation ("commercial use"): e.g. journal can make profit while author doesn't.
- Non-derivatives: no adaptations can be made.
  - → Strong limitation for dataset reuse and/or integration.

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DATACITE METADATA SCHEMA

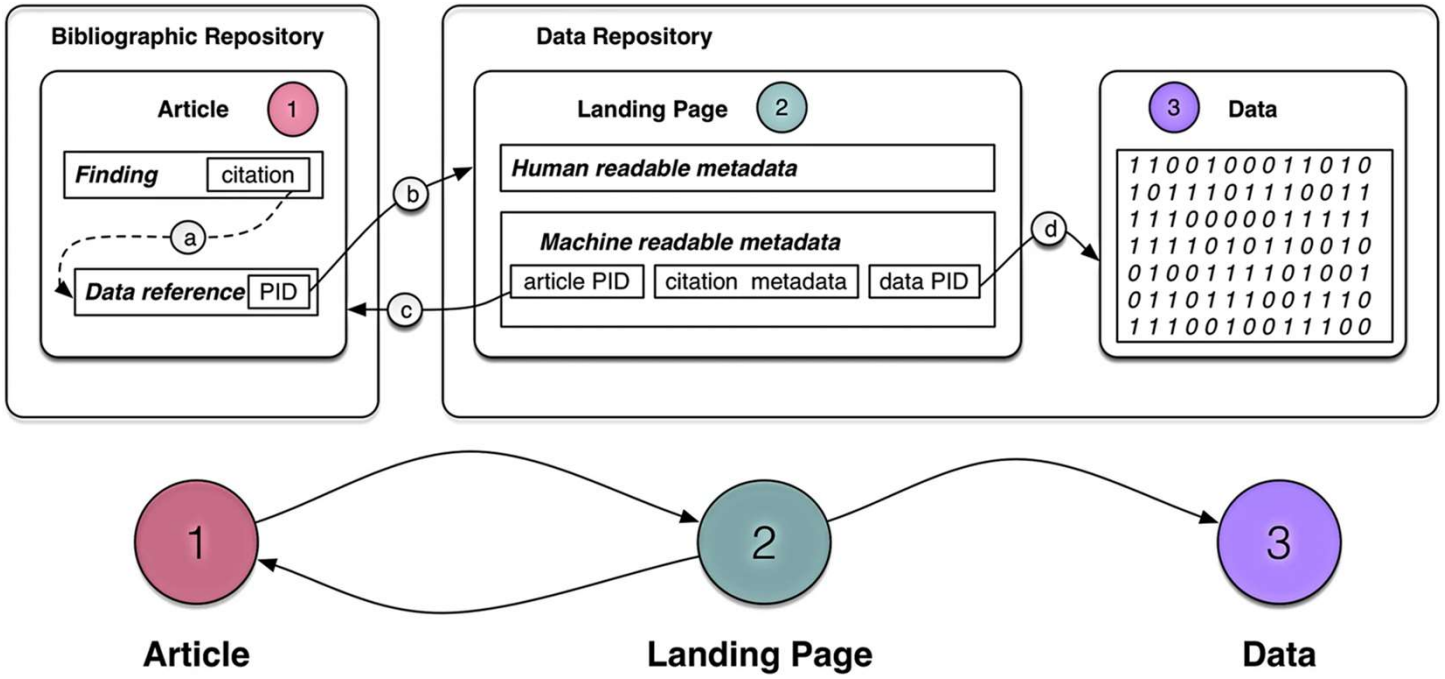
Mandatory	Recommended	Optional
Identifier	Subject	Language
Creator	Contributor	Alternate ID
Title	Date	Size
Publisher	Resource Type	Format
Publication year	Related identifier	Version
	Description	Rights
	GeoLocation	

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LEGITIMATE AND CITABLE PRODUCTS OF RESEARCH

Making your data citable enables you to **claim and receive credit** for producing high-quality datasets and enhances the potential **impact** of your research.

When you **reuse data** from someone else, in turn also **cite these in your publications** to contribute to a culture of data citation.



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Cousijn, H., Kenall, A., Ganley, E. *et al.* A data citation roadmap for scientific publishers. *Sci Data* **5,** 180259 (2018). https://doi.org/10.1038/sdata.2018.259

DOI: MOST COMMONLY USED IDENTIFIER FOR ARTICLES AND DATA

Article

Behavioral Ecology

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Volume 30, Issue 4
July/August 2019

Article Contents

Abstract
Introduction
MATERIAL AND METHODS
Results

Sharing the burden: on the division of parental care and vocalizations during incubation

Marwa M Kavelaars, Luc Lens, Wendt Müller

Behavioral Ecology, Volume 30, Issue 4, July/August 2019, Pages 1062–1068,
https://doi.org/10.1093/beheco/arz049
Published: 12 April 2019 **Article history**

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Abstract

In species with biparental care, individuals only have to pay the costs for their own parental investment, whereas the contribution of their partner comes for free. Each parent hence benefits if its partner works harder, creating an evolutionary conflict of interest. How parents resolve this conflict and how they achieve the optimal division of parental tasks often remains elusive. In this

. . .

Data accessibility: Analyses reported in this article can be reproduced using the data provided by Kavelaars et al. (2019).

. . .

Kavelaars MM, Lens L, Müller W. 2019. Data from: sharing the burden: on the division of parental care and vocalizations during incubation. Dryad Digital Repository. doi: 10.5061/dryad.4h16331.

Data

DRYAD

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Data from: Sharing the burden: on the division of parental care and vocalizations during incubation

Kavelaars, Marwa M., University of Antwerp
Lens, Luc, Ghent University
Müller, Wendt, University of Antwerp
Publication date: March 15, 2019
Publisher: Dryad
https://doi.org/10.5061/dryad.4h16331

Citation

Kavelaars, Marwa M.; Lens, Luc; Müller, Wendt (2019), Data from: Sharing the burden: on the division of parental care and vocalizations during incubation, Dryad, Dataset, https://doi.org/10.5061/dryad.4h16331

Abstract

In species with biparental care, individuals only have to pay the costs for their own parental investment, while the contribution of their partner comes for free. Each parent hence benefits if its partner works harder, creating an evolutionary conflict of interest. How parents resolve this conflict and how they achieve the optimal division of parental tasks often remains elusive. In this study, we investigated whether lesser black-backed gulls (Larus fuscus) divide parental care during incubation equally and whether this correlates with the extent of vocalizations between pair-members during incubation. We then investigated whether pairs showing more evenly distributed incubation behavior had a higher

Data Files

Download dataset
March 15, 2019

Related Works

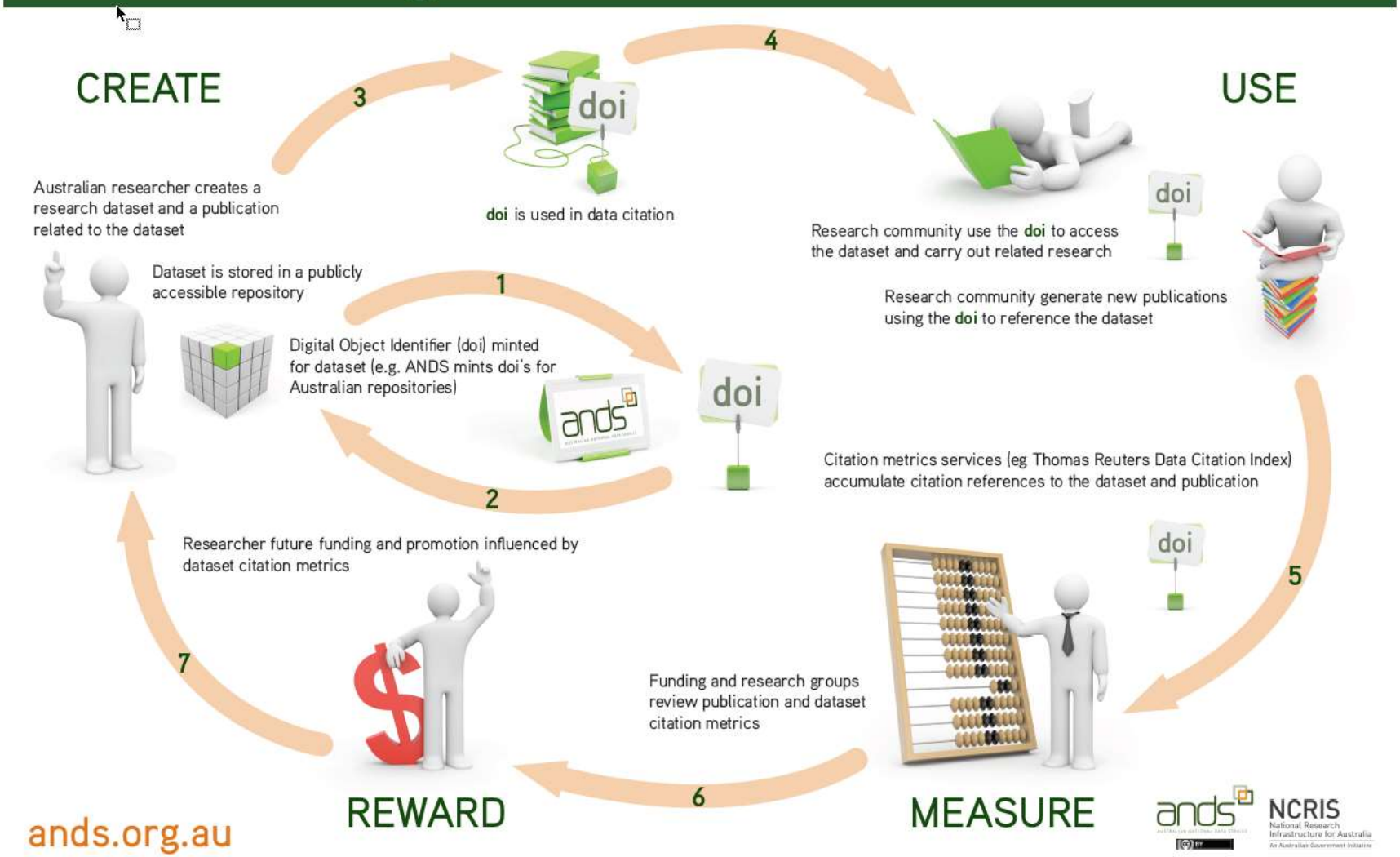
Article
https://doi.org/10.1093/beheco/arz049

Metrics

124 views
14 downloads
1 citations

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Building a Culture of Data Citation



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ands.org.au

NCRIS National Research Infrastructure for Australia

POLL

Are you already taking some of the following actions to increase FAIRness of data?

- I generate documentation during research
- I use standard data formats
- I use standard vocabularies and/or metadata
- I deposit data in a data repository
- I assign a license to my data
- I cite data and/or add a data availability statement in my articles

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FAIR @ UGENT

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FAIR DATA PRINCIPLES



ARDC 2018

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Wilkinson, M., Dumontier, M., Aalbersberg, I. *et al.* The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* **3,** 160018 (2016). https://doi.org/10.1038/sdata.2016.18

WHY DOMAIN-SPECIFIC DATA REPOSITORIES?

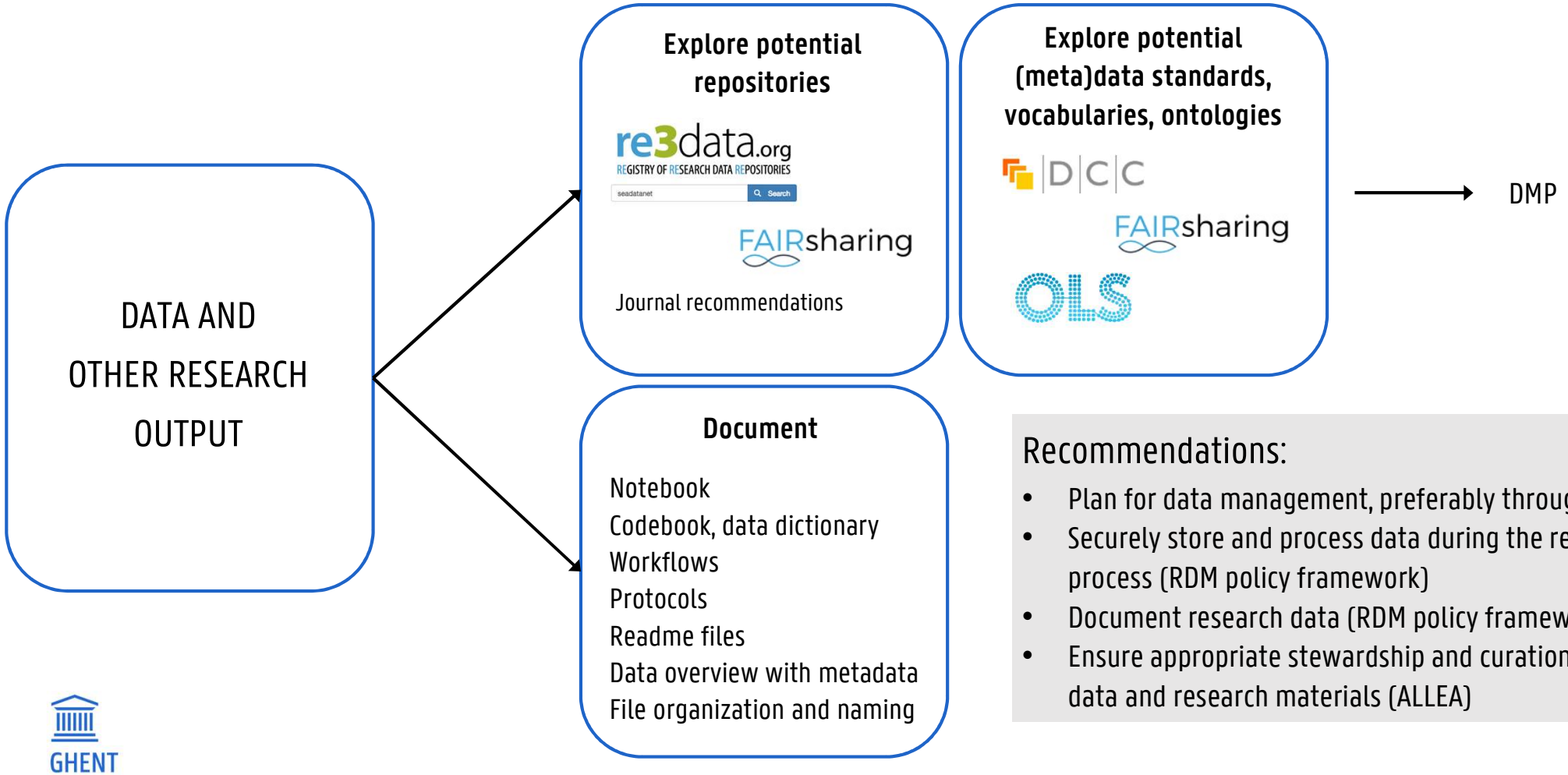
→ Key infrastructures that provide functionalities to make your data FAIR

→ Domain-specific documentation and terminology increase (re-)usability

FAIR principle	In practice	Repository	Researcher
FINDABLE	Assign a persistent identifier (PID)	●	
	Data are described with rich metadata	●	●
	Metadata include the PID, are indexed & searchable online	●	
ACCESSIBLE	(Meta)data retrievable via PID using a standardised communications protocol (e.g. html)	●	
	Authentication and authorisation procedure	●	
INTEROPERABLE	Metadata standards	●	
	Controlled vocabularies	●	●
	References to other (meta)data	●	●
REUSABLE	(Machine readable) license / terms of use	●	●
	Detailed provenance documentation		●
	Domain standards	●	●

Shared responsibilities between repositories and researchers in making data FAIR

MAXIMISING DATA RE-USE DURING RESEARCH



Recommendations:

- Plan for data management, preferably through a DMP
- Securely store and process data during the research process (RDM policy framework)
- Document research data (RDM policy framework)
- Ensure appropriate stewardship and curation of all data and research materials (ALLEA)

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HOW TO SELECT A DATA REPOSITORY?

Recommended repositories

RESEARCH DOMAIN'S CONSENSUS



RESEARCHERS' BEST PRACTICES



JOURNAL RECOMMENDATIONS



data repositories
recommended by Nature

DATA REPOSITORY REGISTRIES

re3data.org
REGISTRY OF RESEARCH DATA REPOSITORIES

FAIRsharing

re3data.org demo

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HOW TO SELECT A DATA REPOSITORY?

Trustworthy repositories



CERTIFIED REPOSITORIES

Core Trust Seal, Nestor Seal
DIN31644 or ISO16363



DOMAIN-SPECIFIC REPOSITORIES

Internationally recognized, commonly used, endorsed by community



FAIR-ENABLING REPOSITORIES

(Generic) repositories with essential characteristics such as authentication & authorization, machine-actionable metadata, long term preservation

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PREPARING DATA FOR SHARING

Without associated information, **research data quickly become useless**.

- Prepare a "data package": all relevant data & documentation needed for reproducibility:
  - experimental method details, raw data files, organized data tables, scripts and code, data visualizations, statistical output…
- Migrate data to recommended, reusable formats
- Organize files in a logical schema
- Provide a readme file



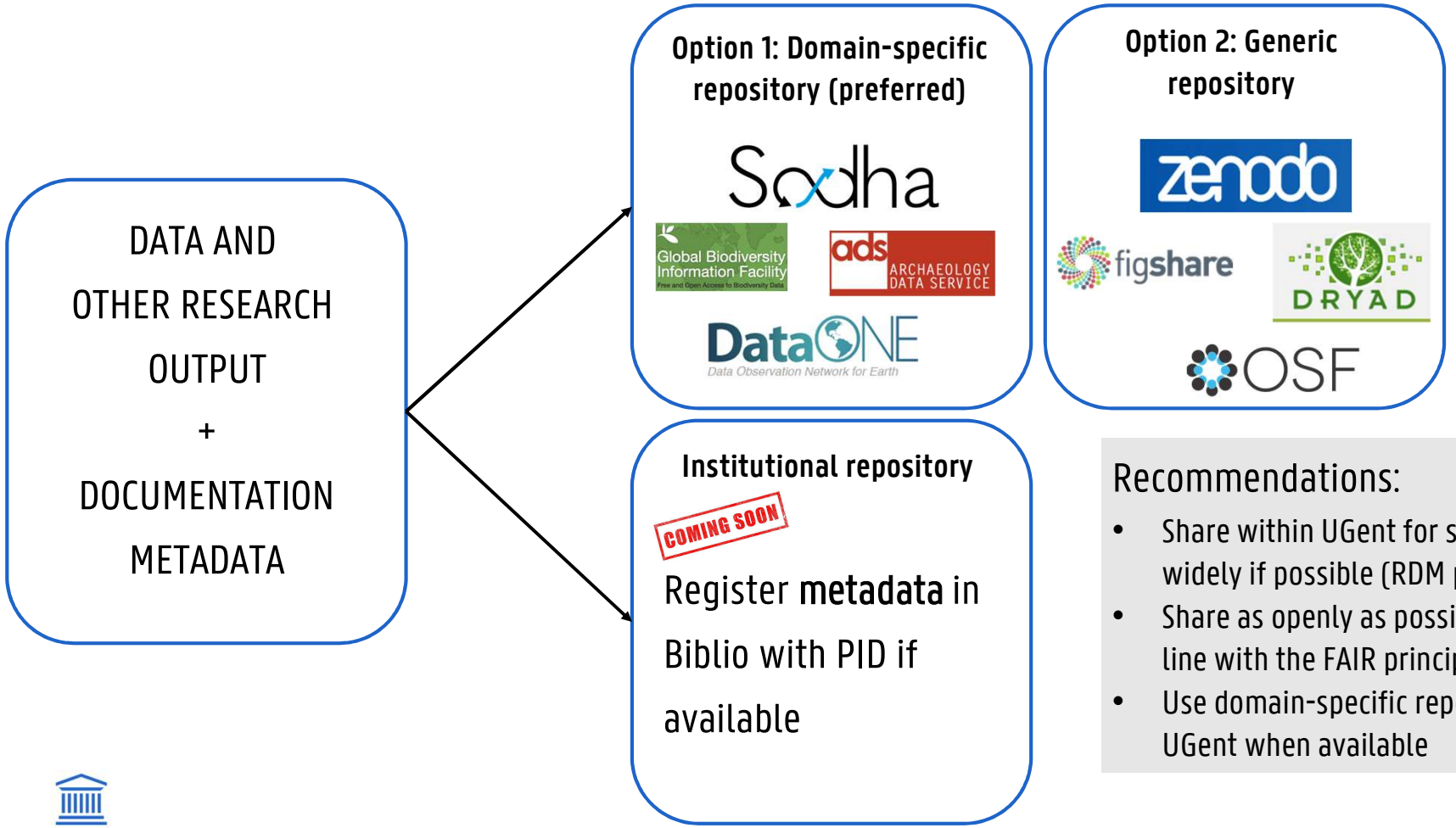
Jørgen Stamp - Digitalbevaring.dk

Established **domain-specific repositories** will usually only accept data that meet their standards for file formats, documentation and metadata, data quality… **Check their requirements** in advance, so you can adequately prepare your data for deposit.

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More info at: https://datadryad.org/stash/best_practices

MAXIMISING DATA RE-USE AFTER RESEARCH



Recommendations:

- Share within UGent for scientific review and more widely if possible (RDM policy framework)
- Share as openly as possible, as closed as necessary, in line with the FAIR principles (ALLEA)
- Use domain-specific repository for sharing outside of UGent when available

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EXAMPLE REPOSITORY

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PANGAEA.

Data Publisher for Earth & Environmental Science

Submit Data

Welcome to PANGAEA® Data Publisher

Our services are generally open for archiving, publishing, and re-usa
PANGAEA is member of the ICSU World Data System.

ALL TOPICS

Search for measurement type, author r

CHEMISTRY (130438)

LITHOSPHERE (130627)

OCEANS (99450)

BIOLOGICAL CLASSIFICAT (33004)

PALEONTOLOGY (25453)

ECOLOGY (17140)

BIOSPHERE (7842)

LAND SURFA (7547)

CRYOSPHERE (1487)

LAKES & RIVERS (688)

HUMAN DIMENSIONS (475)

FISHERIES (278)

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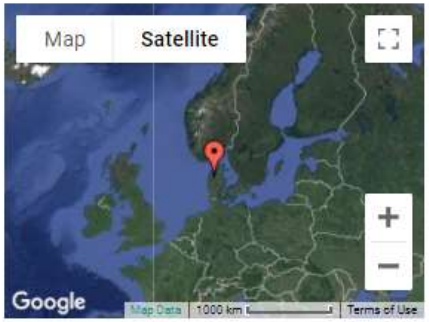




Citation:

Saurel, Camille; Mohn, Christian; Andersen, Kasper; Andersen, Lars; Bak, Finn; Barreau, Pascal; Larsen, Janus; Maar, Marie; Nielsen, Niels Peter; Pastor, Ane; Petersen, Jens Kjerulf (2021): Water current velocity time series from Lovns Bredning, Limfjorden, Denmark, 7-9 March 2017. *PANGAEA*, https://doi.pangaea.de/10.1594/PANGAEA.927177 *(DOI registration in progress)*,

In: Saurel, C et al. (2021): Water currents and light intensity data for estimating sediment plume size and light conditions from mussel dredging in two areas of the Limfjorden, Denmark. *PANGAEA*, https://doi.org/10.1594/PANGAEA.927135



Always quote citation above when using data! You can download the citation in several formats below.

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Abstract:	A 600 kHz ADCP (RDI Workhorse Sentinel) was deployed at each site (station 1 at the Lovns Bredning mooring, station 3 at the Løgstør Bredning mooring) in a bottom-mounted, upward-looking configuration. 30-second ensembles of 3D velocity components were collected along with corresponding records of beam correlation and error velocity. Each vertical profile had a bin size of 0.5 m and the first bin was at 1.59 m above the bottom (blanking distance: 0.88 m). The velocity data time series at each bin were filtered using a 30 minute moving average.
Keyword(s):	ADCP; Limfjorden; mussel dredging; Sediment plumes
Related to:	**Pastor, Ane; Larsen, Janus; Mohn, Christian; Saurel, Camille; Petersen, Jens Kjerulf; Maar, Marie (2020):** Sediment Transport Model Quantifies Plume Length and Light Conditions From Mussel Dredging. *Frontiers in Marine Science*, **7**, https://doi.org/10.3389/fmars.2020.576530
Further details:	Coordinates Lovns Bredning, Limfjorden
Coverage:	*Latitude:* 56.629200 * *Longitude:* 9.172900 *Date/Time Start:* 2017-02-27T15:07:00 * *Date/Time End:* 2017-03-03T11:37:00 *Minimum DEPTH, water:* 0.91 m * *Maximum DEPTH, water:* 3.41 m

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Event(s): **TLi_01** * *Latitude:* 56.629200 * *Longitude:* 9.172900 * *Method/Device:* Multiple investigations (MULT) * *Comment:* average water depth: 4.8 m

Parameter(s):

#	Name	Short Name	Unit	Principal Investigator	Method/Device	Comment
1	DATE/TIME	Date/Time		Mohn, Christian		Geocode – local time (CET)
2	DEPTH, water	Depth water	m	Mohn, Christian		Geocode
3	Current velocity, east-west	Cur vel U	cm/s	Mohn, Christian	Acoustic Doppler Current Profiling (ADCP), RDI Workhorse Sentinel, 600 kHz	
4	Current velocity, north-south	Cur vel V	cm/s	Mohn, Christian	Acoustic Doppler Current Profiling (ADCP), RDI Workhorse Sentinel, 600 kHz	
5	Current velocity, vertical	Cur vel W	cm/s	Mohn, Christian	Acoustic Doppler Current Profiling (ADCP), RDI Workhorse Sentinel, 600 kHz	

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Size: 9315 data points

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CREATE FAIR DATA WITH A REPOSITORY



https://youtu.be/5bJfSuAukTQ

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UGent Data Stewards

RESEARCH DEPARTMENT - UNIVERSITY LIBRARY

rdm.support@ugent.be

@UGentRDM

ugent.be/en/research/datamanagement

Knowledge clip on YouTube: FAIR data principles
https://www.youtube.com/watch?v=2uZxFu9SFi8

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DOCUMENTATION RESOURCES

- READMEs
  - Guide to writing “readme” style metadata
  - README template
  - Documentation methods University of Bath
- Data Dictionaries
  - How to Make a Data Dictionary
  - Data Dictionary Template
- Codebooks
  - Codebook Cookbook
  - Sample Questionnaire with Coding
- Commented Code
  - Coding and Comment Style
- Lab/field Notebook
  - Guide for Taking Field Notes

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CHECKLISTS, VALIDATION TOOLS OR TEMPLATES FOR DATA ENTRY

- Checklists:
  - Minimum information about a genome sequence (MIGS)
- File format validation or curation:
  - Data curation primers
  - fastq format validation
- Data templates:
  - For data collection
  - Top 10 spreadsheet best practices, spreadsheet use cases
  - EarthChem data templates

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